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Internship Name: Machine
Learning

Assignment 1

**Task 1: Descriptive Statistics,
Probability & Combinatorics (Manual)**

1.1 Summary Statistics

Data: 72, 85, 90, 65, 78, 92, 68, 75, 88, 80

$$\text{Mean} = \frac{\sum x}{n}$$

$$\Rightarrow \text{Mean} = \frac{72 + 85 + 90 + 65 + 78 + 92 + 68 + 75 + 88 + 80}{10}$$

$$= \frac{793}{10} = 79.3$$

$$\text{Median} = \left[\frac{n}{2}^{\text{th}} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \right] / 2$$

(No. of terms is odd/even)

Sorted data: 65, 68, 72, 75, 78, 80, 85, 88, 90, 92

$$5^{\text{th}} \text{ value} = 78$$

$$6^{\text{th}} \text{ value} = 80$$

$$\text{Median} = \frac{78 + 80}{2} = 79$$

Mode

 $\Rightarrow$ No mode

Range = Maximum - Minimum
 $\Rightarrow \text{Range} = 92 - 65 = 27$

Variance : $s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$

x	$(x - 79.3)$	$(x - 79.3)^2$
72	-7.3	53.29
85	5.7	32.49
90	10.7	114.49
65	-14.3	204.49
78	-1.3	1.69
92	12.7	161.29
68	-11.3	127.69
75	-4.3	18.49
88	8.7	75.69
80	0.7	0.49

Variance = $53.29 + 32.49 + 114.49 + 204.49 + 1.69 + 161.29 + 127.69 + 18.49 + 75.69 + 0.49$
 $\frac{10-1}{= 87.79}$

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$$\text{Standard Deviation} = \sqrt{\text{variance}}$$

$$\Rightarrow SD = \sqrt{81.79} = 9.37$$

1.2 Probability

Ans. Coin tossed = 3 times
Probability of getting exactly two heads : Favourable outcomes

Total outcomes

Total outcomes : (HHH), (HHT), (HTH), (HTT), (TTT), (THT), (TTH), (THT)

$$\text{Probability} = \frac{3}{8}$$

Ans. No. of red balls = 5

No. of blue balls = 4

No. of green balls = 3

$$P(\text{red}) = \frac{5}{12}$$

$$P(\text{not green}) = \frac{9}{12} = \frac{3}{4}$$

A3. Two dice are rolled together,

Total outcomes:

(1,1)(1,2)(1,3)(1,4)(1,5)(1,6)
(2,1)(2,2)(2,3)(2,4)(2,5)(2,6)
(3,1)(3,2)(3,3)(3,4)(3,5)(3,6)
(4,1)(4,2)(4,3)(4,4)(4,5)(4,6)
(5,1)(5,2)(5,3)(5,4)(5,5)(5,6)
(6,1)(6,2)(6,3)(6,4)(6,5)(6,6)

$$P(\text{Sum} > 8) = \frac{10}{36} = \frac{5}{18}$$

1.3 Permutations and Combinations

A10. Vowels: A, T, E

Treat all the vowels of one block.

Now, 4 consonants are left:

M, C, H, N

These 5 can be arranged in $5! = 120$ ways.

Three vowels themselves can be arranged in $3! = 6$ ways

$$\begin{aligned}\text{Total arrangements} &= 120 \times 6 \\ &= 720 \text{ ways}\end{aligned}$$

A2. No. of ways to choose 2 men: 6C_2
$$= \frac{6!}{2!(6-2)!}$$
$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 4 \times 3 \times 2 \times 1} = 15$$

No. of ways to choose 2 women: 5C_2
$$= \frac{5!}{2!(5-2)!}$$
$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1} = 10$$

Total no. of ways to select 4 members
 $= 15 \times 10 = 150$ ways

1.4 Discrete Probability Distribution

A- No. of trials = 5
Probability of getting a head
 $\Rightarrow p = 0.6$
Probability of getting a tail
 $\Rightarrow q = 1 - 0.6 = 0.4$

Binomial Probability formula:

$$P(X=k) = {}^n C_k p^k q^{n-k}$$

$$\begin{aligned} (a) P(X=3) &= {}^5 C_3 p^3 q^{5-3} \\ &= \frac{5!}{3!(5-3)!} (0.6)^3 (0.4)^2 \\ &= \frac{5 \times 4 \times 3!}{3! \times 2 \times 1} \times (0.6)^3 \times (0.4)^2 \\ &= 0.3456 \end{aligned}$$

$$\begin{aligned} (b) P(X \geq 4) &= P(X=4) + P(X=5) \\ &= {}^5 C_4 p^4 q^{5-4} + {}^5 C_5 p^5 q^{5-5} \\ &= \frac{5!}{4!(5-4)!} p^4 q + \frac{5!}{5!(5-5)!} p^5 q^0 \\ &= \frac{5 \times 4!}{4!} (0.6)^4 (0.4) + 1 \times (0.6)^5 \times 1 \\ &= 0.2592 + 0.07776 \\ &= 0.33696 \end{aligned}$$

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Task 2: Python, Numpy & Pandas

2.1 Verify Statistics with Numpy

```
[1]: import numpy as np
```

```
[2]: marks=np.array([72, 85, 90, 65, 78, 92, 68, 75, 88, 80])
print("Mean:",np.mean(marks))
print("Median:",np.median(marks))
print("Range:",np.max(marks)-np.min(marks))
print("Variance(sample):",np.var(marks,ddof=1))
print("Std Deviation(sample):",np.std(marks, ddof=1))
```

Mean: 79.3
Median: 79.0
Range: 27
Variance(sample): 87.78888888888889
Std Deviation(sample): 9.369572502995474

2.2 Pandas Dataframe Operations

```
[3]: import pandas as pd
```

```
[4]: df=pd.read_csv("D:/ML Internship/employees.csv")
df
```

```
[4]:
```

	emp_id	name	department	age	salary	experience_yrs
0	E01	Aarav	IT	28	55000	3

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2.2 Pandas Dataframe Operations

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[3]: import pandas as pd
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[4]: df=pd.read_csv("D:/ML Internship/employees.csv")
df
```

```
[4]:
```

	emp_id	name	department	age	salary	experience_yrs
0	E01	Aarav	IT	28	55000	3
1	E02	Bhavya	HR	32	48000	5
2	E03	Chirag	IT	26	52000	2
3	E04	Diya	Finance	35	72000	8
4	E05	Eshan	IT	30	60000	4
5	E06	Farah	Sales	29	45000	3
6	E07	Gaurav	Finance	40	85000	12
7	E08	Hina	HR	27	42000	2
8	E09	Ishan	Sales	33	55000	6
9	E10	Jiya	IT	38	78000	10

```
[5]: print(df.head())
```

	emp_id	name	department	age	salary	experience_yrs
0	E01	Aarav	IT	28	55000	3
1	E02	Bhavya	HR	32	48000	5


```
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JupyterLab Python 3 (ipykernel)

[5]: print(df.head())
emp_id  name department  age  salary  experience_yrs
0    E01  Aarav      IT    28   55000             3
1    E02  Bhavya     HR    32   48000             5
2    E03  Chirag     IT    26   52000             2
3    E04  Diya      Finance 35   72000             8
4    E05  Eshan      IT    30   60000             4

[6]: print(df.describe())
count    age      salary  experience_yrs
count  10.000000  10.000000  10.000000
mean    31.800000  59200.000000  5.500000
std     4.709329  14505.171492  3.472111
min     26.000000  42000.000000  2.000000
25%     28.250000  49000.000000  3.000000
50%     31.000000  55000.000000  4.500000
75%     34.500000  69000.000000  7.500000
max     40.000000  85000.000000  12.000000

[7]: print(df.groupby("department")["salary"].mean())
department
Finance    78500.0
HR         45000.0
IT         61250.0
Sales      50000.0
Name: salary, dtype: float64

[8]: print(df[df["experience_yrs"]>5])
emp_id  name department  age  salary  experience_yrs
3    E04  Diya      Finance 35   72000             8
6    E07  Gaurav    Finance 40   85000            12
8    E09  Ishan     Sales  33   55000             6
```

```
Name: salary, dtype: float64

[8]: print(df[df["experience_yrs"]>5])
emp_id  name department  age  salary  experience_yrs
3    E04  Diya      Finance 35   72000             8
6    E07  Gaurav    Finance 40   85000            12
8    E09  Ishan     Sales  33   55000             6
9    E10  Jiya      IT    38   78000            10

[9]: print(df.sort_values(by="salary",ascending=False).head(3))
emp_id  name department  age  salary  experience_yrs
6    E07  Gaurav    Finance 40   85000            12
9    E10  Jiya      IT    38   78000            10
3    E04  Diya      Finance 35   72000             8

[10]: df["salary_per_year_exp"]=df["salary"]/df["experience_yrs"]
print(df)
emp_id  name department  age  salary  experience_yrs  salary_per_year_exp
0    E01  Aarav      IT    28   55000             3    18333.333333
1    E02  Bhavya     HR    32   48000             5     9600.000000
2    E03  Chirag     IT    26   52000             2    26000.000000
3    E04  Diya      Finance 35   72000             8     9000.000000
4    E05  Eshan      IT    30   60000             4    15000.000000
5    E06  Farah     Sales  29   45000             3    15000.000000
6    E07  Gaurav    Finance 40   85000            12     7083.333333
7    E08  Hina      HR    27   42000             2    21000.000000
8    E09  Ishan     Sales  33   55000             6     9166.666667
9    E10  Jiya      IT    38   78000            10     7800.000000

Task 3: Inferential Statistics+Visualization
```

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Task 3: Inferential Statistics+Visualization

3.1 Hypothesis Testing

```
[17]: from scipy import stats
data=[7.5, 8.2, 7.8, 9.0, 8.5, 7.2, 8.8, 9.1, 7.4, 8.0]
t_stat,p_value=stats.ttest_1samp(data,popmean=8)
print("t-statistic=",t_stat)
print("p-value=",p_value)
```

t-statistic= 0.6939521309482124
p-value= 0.505227742792751

Since:

1. p-value=0.505>0.05, and
2. t-calculated is less than t-tabulated so, null hypothesis is accepted.

Final Conclusion:

At the 5% significance level, there is insufficient evidence to conclude that the average daily working hours of employees differ from 8 hours. Therefore, the company's claim is accepted based on the sample data.

3.3 Matplotlib Visualisation

```
[12]: import matplotlib.pyplot as plt
```

```
[13]: df
```

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3.3 Matplotlib Visualisation

```
[12]: import matplotlib.pyplot as plt
```

```
[13]: df
```

```
[13]:
```

	emp_id	name	department	age	salary	experience_yrs	salary_per_year_exp
0	E01	Aarav	IT	28	55000	3	18333.333333
1	E02	Bhavya	HR	32	48000	5	9600.000000
2	E03	Chirag	IT	26	52000	2	26000.000000
3	E04	Diya	Finance	35	72000	8	9000.000000
4	E05	Eshan	IT	30	60000	4	15000.000000
5	E06	Farah	Sales	29	45000	3	15000.000000
6	E07	Gaurav	Finance	40	85000	12	7083.333333
7	E08	Hina	HR	27	42000	2	21000.000000
8	E09	Ishan	Sales	33	55000	6	9166.666667
9	E10	Jiya	IT	38	78000	10	7800.000000

```
[14]: avg_salary=df.groupby("department")["salary"].mean()
avg_salary
```

```
[14]: department
Finance    78500.0
HR         45000.0
IT         61250.0
```



